

LIN LIU

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EDUCATION

Ph.D. in Geophysics , University of Colorado at Boulder, USA	2011
B.Sc. in Geophysics , Wuhan University, China	2005

RESEARCH INTERESTS

Cryosphere and Polar Sciences	Geodesy and Geophysics	Remote Sensing
Planetary Geomorphology	Artificial Intelligence for Earth and Environmental Sciences	

PROFESSIONAL EXPERIENCE

Faculty Member , The Chinese University of Hong Kong (CUHK)	2014–present
• Professor, Department of Earth and Environmental Sciences, Faculty of Science • Research Fellow, Institute of Environment, Energy and Sustainability (since 2014) • Research Fellow, Institute of Space and Earth Information Science (since 2014) • Associate Professor (2020–2026) • Assistant Professor (2014–2019)	
Invited Professorship , Université Grenoble Alpes	2025
Scientific Team Member , China's 41st Expedition to Antarctica	2024–2025
Visiting Scholar , University of Utah	2024
Visiting Scholar , Stanford University	2014–2019
George Thompson Postdoctoral Fellow , Stanford University	2011–2013
Research Assistant , University of Colorado at Boulder	2006–2011

HONORS AND AWARDS (SELECTED)

Invited Professor , Grenoble INP–Université Grenoble Alpes	2025
Distinguished Faculty Stay Award , Department of Geology and Geophysics, University of Utah	2024
Exemplary Teaching Award , Faculty of Science, CUHK	2016, 2023
John Wahr Early Career Award , American Geophysical Union	2021
George Thompson Postdoctoral Fellowship , Department of Geophysics, Stanford University	2011
NASA Earth and Space Science Fellowship , National Aeronautics and Space Administration	2008
CIRES Graduate Research Assistant Fellowship , University of Colorado	2006
First Prize, Outstanding Bachelor's Thesis of Hubei Province	2005
Xia Jianbai Scholarship in Geophysics and Geodesy , Wuhan University	2003

TEACHING EXPERIENCE

Course Instructor , CUHK	
• Applied Geophysics (EESC4110 & 5110) • Remote Sensing: Principles and Applications (EESC4540 & 5540) • Introduction of AI for Science (AISC5030, co-teaching) • Geomorphology (EESC4130, co-taught with Tammy Tam) • Earth System Science (ESGS5001) • Exploring the Earth System (ESSC1000) • Solid Earth Dynamics (ESSC2010) • Geodesy and Gravity (ESSC4601 & EASC5601) • Research Frontiers in Earth and Atmospheric Sciences (EASC5002) • Engineering Geology and Applied Geophysics (co-taught with Teng-fong Wong) • Hydrogeology (ESSC3220, co-taught with Teng-fong Wong)	
	2017–present
	2014–present
	2026–present
	2025–present
	2021–present
	2016–2023
	2014–2023
	2020, 2022
	2014–2020
	2016
	2014
Teaching Assistant , Department of Physics, University of Colorado	2005–2006

Ongoing Projects

1. **Seasonal and inter-annual evolution of ponded crevasses on the Greenland Ice Sheet**
PI, Hong Kong Research Grants Council (RGC) General Research Fund 2027–
2. **Carbon release from abrupt thaw of frozen ground in the Arctic and its impacts on carbon neutrality**
PI, Hong Kong RGC GRF 2026–
3. **Unveiling climate-driven biogeochemical changes on the Antarctic Peninsula**
Project Coordinator, CUHK Science Faculty Collaborative Research Impact Matching Scheme 2025–
4. **Cooperative research on subglacial geological environment detection technology in the Larsemann Hills area, East Antarctica**
Co-Investigator, Ministry of Science and Technology of China–Key Technologies R&D Program 2025–
5. **Multi-sensor monitoring, geophysical interpretation and prediction of sea level rise in Hong Kong**
Co-PI, Hong Kong RGC Collaborative Research Fund 2024–
6. **A comparative study on the evolution of habitability between Earth and Mars**
Co-Investigator, Ministry of Science and Technology of China–Key Technologies R&D Program 2024–

Completed Projects

1. **Thermokarst Landforms on the Qinghai-Tibet Plateau: Spatio-temporal Evolution and Future Changes**
Co-PI, National Natural Science Foundation of China (NSFC)/RGC Joint Research Scheme 2022–2025
2. **Characterization of ancient lake basins on Mars using advanced topographic modeling and innovative spectroscopic techniques** *Co-PI*, Hong Kong RGC Collaborative Research Fund 2022–2025
3. **Deep-learning-based mapping of rock glaciers on the Qinghai-Tibet Plateau**
PI, RGC General Research Fund (GRF) 2022–2024
4. **Rock glacier distribution across the Hindu Kush Himalaya and the French Alps based on remote sensing and deep learning** *Co-PI*, PROCORE-France/Hong Kong Joint Research Scheme 2023–2025
5. **Hydrological significance of rock glaciers: potential water resources in a warming climate**
Co-PI, CUHK–University of Exeter Joint Centre for Environmental Sustainability & Resilience 2021–2024
6. **DeepThaw: Deep-learning-based mapping of thermokarst landforms on the Tibetan Plateau**
PI, RGC GRF 2020–2023
7. **GNSS-RECIPE: Global Navigation Satellite System Reflectometry Studies of Elevation Changes in Permafrost Areas** *PI*, RGC GRF 2019–2021
8. **Kinematics and dynamics of active rock glaciers in western China** *PI*, RGC GRF 2018–2021
9. **Earth observation to investigate the characteristics and changes of the cryosphere in High Mountain Asia**
Co-Investigator, NRSCC/ESA Dragon 4 Programme 2016–2020
10. **Investigation of characteristics and mechanism of earthquakes associated with the Hutubi gas reservoir**
Co-Investigator (with T.-f. Wong as *PI*), NSFC/RGC Joint Research Scheme 2016–2019
11. **Radar remote sensing investigations on thermokarst dynamics on the Qinghai-Tibet Plateau, China**
PI, RGC GRF 2016–2019
12. **Mass balance of Greenland outlet glaciers: Non-secular variations from space geodetic measurements**
PI, RGC Early Career Scheme Grant 2015–2018
13. **The combined use of L- and P-band radar to retrieve active layer thickness over Arctic permafrost**
PI, CUHK Direct Grant for Research 2017–2018
14. **Synergistic investigations of surface deformation in permafrost areas using field and remote sensing observations** *Co-PI*, Hong Kong RGC Germany/Hong Kong Joint Research Scheme 2016–2017
15. **Radar Remote Sensing and Field Investigation of Permafrost Changes in Svalbard**
PI, CUHK Direct Grant for Research 2014–2015

16. **Remotely-Sensed Active Layer Thickness (ReSALT) product derived from InSAR data over North American Arctic regions** *Co-Investigator*, NASA Terrestrial Ecology Program 2013–2016
17. **Collaborative Research: Exploring the Dynamics of the Active Layer and Near-surface Permafrost across the North Slope of Alaska** *Co-Investigator*, NSF Arctic Natural Science Program 2012–2015

PUBLICATIONS

Annotations: graduate students, postdocs[×], undergraduate^{*}, and visiting students[◇]

1. Peacey, M., Harrison, S., Irvine-Fynn, T., Hu, Y., Anderson, K., Leng, R., Sun, Z., and Liu, L. (2026), First empirical assessment of ice content from a Himalayan rock glacier, *Geophysical Research Letters*, in press.
2. Xia, Z., Zhao, Z., Liu, L., Huang, L., Gascoin, S., and Lara, M. (2026), Life cycle of retrogressive thaw slumps in a warmer and wetter climate on the Central Qinghai-Tibet Plateau, *npj Natural Hazards*, <https://doi.org/10.1038/s44304-026-00258-9>.
3. Guo, X., Wang, X., Zhang, S., Sun, Z., Liu, L., Liu, G., and Bolch, T. (2026), Rock glaciers of the Southeastern Tibetan Plateau: A large-scale inventory derived from deep learning and high-resolution imagery, *Scientific Data*, <https://doi.org/10.1038/s41597-026-08019-z>.
4. Jia, Y., Zhang, C., Mu, M., Du, X., Song, J., Wei, Y., Ge, Y., Li, K., Liu, H., Lei, P., Peng, X., Xia, Z., Huang, L., Liu, R., Wangchuk, S., Yang, Y., Liu, L., and Mu, C. (2026), Abrupt permafrost thaw drives exceptional carbon release across the Tibetan Plateau, *Nature Communications*, 17, 8050, <https://doi.org/10.1038/s41467-026-74850-y>.
5. Chang, T., Yi, Y., Furuya, M., Jiang, H., Che, T., Ran, Y., Liu, L., and Li, R. (2026), Long-term InSAR and stream-flow recession analysis reveal accelerated permafrost degradation in the mining area of Qilian Mountain, *The Cryosphere*, 20, 3933–3957, <https://doi.org/10.5194/tc-20-3933-2026>.
6. Xia, Z., Liu, L., Nitze, I., Nesterova, N., van der Sluijs, J., Zhu, X., Wu, T., Ermokhina, K., Hall, E., Khairullin, R., Khomutov, A., and Lara, M. (2026), Vegetation recovery following retrogressive thaw slumps across northern tundra regions, *Nature Climate Change*, 16, 606–612, <https://www.nature.com/articles/s41558-026-02603-2>.
7. Jiang, H., Qiang, D., Yi, Y., Goetz, S., Berner, L., Rogers, B., Liu, L., He, M., Chen, D., Ma, K. (2026), Divergent responses of vegetation growth to recent warming and abrupt permafrost thaw across the Qinghai-Tibet plateau, *Advances in Climate Change Research*, <https://doi.org/10.1016/j.accre.2026.06.002>.
8. Sun, Z., Liu, L., and Bolch, T. (2026), Uncovering controlling factors on rock glacier velocities in the Pamir-Karakoram-Kunlun region using explainable machine learning, *PNAS Nexus*, 5(5), pgag177, <https://doi.org/10.1093/pnasnexus/pgag177>.
9. Du, X., Zhang, C., Jia, Y., Wei, Y., Song, J., Mu, M., Peng, X., Li, Z., Zhang, P., Liu, L., Xia, Z., Liu, R., Wangchuk, S., Zhang, X., Hjort, J., and Mu, C. (2026), Permafrost degradation amplifies infrastructure-related inequalities across the Qinghai-Tibet Plateau, *Global Environmental Change*, 99, 103170, <https://doi.org/10.1016/j.gloenvcha.2026.103170>.
10. Peng, X., Tian, W., Luo, H., Xia, Z., Frauenfeld, O., Mu, C., Liu, L., Luo, J., Huang, L., Zhao, Q., and Yin, Y. (2026), Underestimated small thermokarst lakes of the Qinghai-Tibet Plateau and their carbon emission potential, *Global Planetary Change*, 261, 105411, <https://doi.org/10.1016/j.gloplacha.2026.105411>.
11. Ma, J., Wang, G., Song, C., Sun, S., Li, J., Deng, Z., Kang, W., Xia, Z., and Liu, L. (2026), Integrating time-series analysis and deep learning methods to reconstruct the long-term retrogressive thaw slumps dynamics in the Qinghai-Tibet Plateau, *GIScience & Remote Sensing*, 63, 2617753, <https://doi.org/10.1080/15481603.2026.2617753>.
12. Lyu, W., Che, Y., Cao, Y., Wang, S., Liu, L., Ma, X., Hu, F., and Pan, Y. (2026), Scalable rain-snow discrimination in glacierized regions using tower observations and time-lapse imagery, *Advances in Climate Change Research*, 17(2), 284–296, <https://doi.org/10.1016/j.accre.2025.12.007>.
13. Maier, K., Xia, Z., Liu, L., Lara, M., van der Sluijs, J., Bernhard, P., and Hajnsek, I., Quantifying retrogressive thaw slump mass wasting and carbon mobilisation on the Qinghai-Tibet Plateau using multi-modal remote sensing, *The Cryosphere*, 19, 4855–4873, <https://doi.org/10.5194/tc-19-4855-2025>.
14. Sun, Z., Liu, L., Fan, C., Hu, Y.[×], Baldacchino, F., Bhattacharya, A., Wood, E., and Bolch, T. (2025), Unveiling large-scale velocity characteristics of rock glaciers in the Tibet-Pamir-Karakoram region using InSAR, *International Journal of Applied Earth Observation and Geoinformation*, 142, 104733, <https://doi.org/10.1016/j.jag.2025.104733>.

15. Xu, X.[×], **Liu, L.**, Huang, L., Hu, Y.[×], Zhang, G., Racoviteanu, A., Liu, E., and Chan, Y.* (2025), Contrasting short-term dynamics of supraglacial ponds along the Hindu Kush-Himalaya revealed by PlanetScope imagery and deep learning, *Global and Planetary Change*, 253, 104949, <https://doi.org/10.1016/j.gloplacha.2025.104949>.
16. Chen, M.[◇], Xu, G., and **Liu, L.** (2025), Estimation of 3D complex deformation of surface-rupture earthquake with automated fault trace identification, *IEEE Transactions on Geoscience and Remote Sensing*, 63, 5213013, <https://doi.org/10.1109/TGRS.2025.3573809>.
17. Liu, S., Zhao, L., Wang, L., **Liu, L.**, Zou, D., Hu, G., Sun, Z., Zhang, Y., Chen, W., Wang, X., Wang, M., Zhou, H., Qiao, Y. (2025), Ground surface deformation in permafrost region on the Qinghai-Tibet Plateau: A review, *Earth-Science Reviews*, 105109, <https://doi.org/10.1016/j.earscirev.2025.105109>.
18. Fan, C.[◇], **Liu, L.**, Zhao, Z.[×], and Mu, C. (2025), Pronounced underestimation of surface deformation due to unwrapping errors over Tibetan Plateau permafrost by Sentinel-1 InSAR: Identification and correction, *Journal of Geophysical Research: Earth Surface*, 130, e2024JF007854, <https://doi.org/10.1029/2024JF007854>.
19. Fan, C.[◇], Mu, C., **Liu, L.**, Zhang, T., Jia, S., Wang, S., Sun, W., and Zhao, Z.[×] (2025), Time-Series models for ground subsidence and heave over permafrost in InSAR Processing: A comprehensive assessment and new improvement, *ISPRS Journal of Photogrammetry and Remote Sensing*, 222, 167–185, <https://doi.org/10.1016/j.isprsjprs.2025.02.019>.
20. Hu, Y.[×], Arenson, L., Barboux, C., Bodin, X., Cicoira, A., Delaloye, R., Gärtner-Roer, I., Kääb, A., Kellerer-Pirklbauer, A., Lambiel, C., **Liu, L.**, Pellet, C., Rouyet, L., Schoeneich, P., Seier, G., and Strozzi, T. (2025), Rock Glacier Velocity: An Essential Climate Variable quantity for permafrost, *Reviews of Geophysics*, 63, e2024RG000847, <https://doi.org/10.1029/2024RG000847>. **Highly-cited Paper**
21. Zhang, Z., Jin, Q., **Liu, L.**, Wang, M., and Zhang, X. (2025), Improved ALT retrieval in the Yellow River source region using time-series InSAR and multilayer soil moisture modeling, *Journal of Hydrology*, 654, 132847, <https://doi.org/10.1016/j.jhydrol.2025.132847>.
22. Streletskiy, D., Maslakov, A., Grosse, G., Shiklomanov, N., Farquharson, L., Zwieback, S., Iwahana, G., Bartsch, A., **Liu, L.**, Strozzi, T., Lee, H., and Debolskiy, M. (2025), Thawing permafrost is subsiding in the Northern Hemisphere—review and perspectives, *Environmental Research Letters*, 20, 013006, <https://doi.org/10.1088/1748-9326/ada2ff>. **Highly-cited Paper**
23. Sun, Z., Hu, Y.[×], Racoviteanu, A., **Liu, L.**, Harrison, S., Wang, X., Cai, J., Guo, X., He, Y., and Yuan, H. (2024), TPRoGI: a comprehensive rock glacier inventory for the Tibetan Plateau using deep learning, *Earth System Science Data*, 16, 5703–5721, <https://doi.org/10.5194/essd-16-5703-2024>.
24. Ran, J., Ditmar, P., van den Broeke, M., **Liu, L.**, Klees, R., Khan, S. A., Moon, T., Li, J., Bevis, M., Zhong, M., Fettweis, X., Liu, J., Noël, B., Shum, C. K., Chen, J., Jiang, L., and van Dam, T. (2024), Vertical bedrock shifts reveal summer water storage in Greenland ice sheet, *Nature*, 635, 108–113, <https://doi.org/10.1038/s41586-024-08096-3>.
25. Chen, J., Zhang, J., Wu, T., **Liu, L.**, Zhang, F., Hao, J., Huang, L., Wu, X., Wang, P., Xia, Z., Zhu, X., and Lou P. (2024), Elevation-dependent shift of landslide activity in mountain permafrost regions of the Qilian Mountains, *Advances in Climate Change Research*, 15(6), 1067–1077, <https://doi.org/10.1016/j.accres.2024.11.003>.
26. Xia, Z., **Liu, L.**, Mu, C., Peng, X., Zhao, Z., Huang, L., Luo, J., and Fan, C. (2024), Widespread and rapid activities of retrogressive thaw slumps on the Qinghai-Tibet Plateau from 2016 to 2022, *Geophysical Research Letters*, 51, e2024GL109616, <https://doi.org/10.1029/2024GL109616>.
27. Xu, X.[×], **Liu, L.**, Huang, L., and Hu, Y.[×] (2024), Combined Use of multi-source satellite imagery and deep learning for automated mapping of glacial lakes in the Bhutan Himalaya, *Science of Remote Sensing*, 10, 100157, <https://doi.org/10.1016/j.srs.2024.100157>.
28. Zwieback, S., **Liu, L.**, Rouyet, L., Short, N., and Strozzi, T. (2024), Advances in InSAR analysis of permafrost terrain, *Permafrost and Periglacial Processes*, 35, 544–556, <https://doi.org/10.1002/ppp.2248>. **Top-cited Paper**
29. Tang, Q., Zhang, G., Yao, T., Wieland, M., **Liu, L.**, and Kaushik, S. (2024), Automatic extraction of glacial lakes from Landsat imagery using deep learning across the Third Pole region, *Remote Sensing of Environment*, 315, 114413, <https://doi.org/10.1016/j.rse.2024.114413>.
30. Ye, Q., Wang, Y., **Liu, L.**, Guo, L., Zhang, X., Dai, L., Zhai, L., Hu, Y., Ali, N., Ji, X., Ran, Y., Qiu, Y., Shi, J., Che, T., Wang, N., Li, X., and Zhu, L. (2024), Remote sensing and modeling of the cryosphere in High Mountain Asia: A

multidisciplinary review, *Remote Sensing*, 16, 1709, <https://doi.org/10.3390/rs16101709>.

31. Ran, J., **Liu, L.**, Zhang, G., Shum, C.K., Qiu, J., Hu, R., Li, J., Peng, J., Hwang, C., Luan, Y., Sun, Y., Xu, M., Chen, D., Ding, J., and Zhong, Y. (2024), Contrasting lake changes in Tibet revealed by recent multi-modal satellite observations, *Science of The Total Environment*, 908, 168342, <https://doi.org/10.1016/j.scitotenv.2023.168342>.
32. Hu, Y.[×], **Liu, L.**, Huang, L., Zhao, L., Wu, T., Wang, X., and Cai, J. (2023), Mapping and characterizing rock glaciers in the arid Western Kunlun Mountains supported by InSAR and deep learning, *Journal of Geophysical Research: Earth Surface*, 128, e2023JF007206, <https://doi.org/10.1029/2023JF007206>.
33. Hu, Y.[×], Harrison, S., **Liu, L.**, and Wood, J. L. (2023), Modelling rock glacier ice content based on InSAR-derived velocity, Khumbu and Lhotse valleys, Nepal, *The Cryosphere*, 17, 2305–2321, <https://doi.org/10.5194/tc-17-2305-2023>.
34. Luo, X., Hu, Z., and **Liu, L.** (2023), Investigating the seasonal dynamics of surface water over the Qinghai-Tibet Plateau using Sentinel-1 imagery and a novel gated multiscale ConvNet, *International Journal of Digital Earth*, 16(1), 1373–1395, <https://doi.org/10.1080/17538947.2023.2198266>.
35. Wu, Z., Ma, P., Zheng, Y., Gu, F., **Liu, L.**, and Lin, H. (2023), Automatic detection and classification of land subsidence in deltaic metropolitan areas using distributed scatterer InSAR and Oriented R-CNN, *Remote Sensing of Environment*, 290, 113545, <https://doi.org/10.1016/j.rse.2023.113545>.
36. Xia, Z., Huang, L.[×], Fan, C., Jia, S., Lin, Z., **Liu, L.**, Luo, J., Niu, F., and Zhang, T. (2022), Retrogressive thaw slumps along the Qinghai-Tibet Engineering Corridor: A comprehensive inventory and their distribution characteristics, *Earth System Science Data*, 14, 3875–3887, <https://doi.org/10.5194/essd-14-3875-2022>.
37. Chen, J., Wu, T., **Liu, L.**, Gong, W., Zwieback, S., Zou, D., Zhu, X., Hu, G., Du, E., Wu, X., Li, R., and Yang, S. (2022), Increased water content in the active layer revealed by regional-scale InSAR and independent component analysis on the central Qinghai-Tibet Plateau, *Geophysical Research Letters*, 49, e2021GL097586, <https://doi.org/10.1029/2021GL097586>.
38. Bolch, T., Yao, T., Bhattacharya, A., Hu, Y., King, O., **Liu, L.**, Pronk, J. B., Rastner, P., and Zhang, G. (2022), Earth observation to investigate occurrence, characteristics and changes of glaciers, glacial lakes and rock glaciers in the Poiqu River Basin (Central Himalaya), *Remote Sensing*, 14, 1927, <https://doi.org/10.3390/rs14081927>.
39. Lai, Y., Zhang, B., Yao, Y., **Liu, L.**, Yan, X., He, Y., and Ou, S. (2022), Reconstructing the data gap between GRACE and GRACE Follow-On at the basin scale using artificial neural network, *Science of The Total Environment*, 823, 153770, <https://doi.org/10.1016/j.scitotenv.2022.153770>.
40. Chen, J., Wu, T., Zou, D., **Liu, L.**, Wu, X., Gong, W., Zhu, X., Li, R., Hao, J., Hu, G., Pang, Q., Zhang, J., and Yang, S. (2022), Magnitudes and patterns of large-scale permafrost ground deformation revealed by Sentinel-1 InSAR on the central Qinghai-Tibet Plateau, *Remote Sensing of Environment*, 268, 112778, <https://doi.org/10.1016/j.rse.2021.112778>.
41. Wang, X., **Liu, L.**, Hu, Y., Wu, T., Zhao, L., Liu, Q., Zhang, R., Zhang, B., and Liu, G. (2021), Multi-decadal geomorphic changes of a low-angle valley glacier in East Kunlun Mountains: remote sensing observations and detachment hazard assessment, *Natural Hazards and Earth System Sciences*, 21, 2791–2810, <https://doi.org/10.5194/nhess-21-2791-2021>.
42. Jia, S., Zhang, T., Fan, C., **Liu, L.**, and Shao, W. (2021), Research progress of InSAR technology in permafrost, *Advances in Earth Science* (in Chinese), 36(7), 694–711, <https://doi.org/10.11867/j.issn.1001-8166.2021.055>.
43. Rouyet, L., **Liu, L.**, Strand, S. M., Christiansen, H. H., Lauknes, T. R., and Larsen, Y. (2021), Seasonal InSAR displacements documenting the active layer freeze and thaw progression in central-western Spitsbergen, Svalbard, *Remote Sensing*, 13, 2977, <https://doi.org/10.3390/rs13152977>.
44. Ran, J., Ditmar, P., **Liu, L.**, Xiao, Y., Klees, R., and Tang, X. (2021), Analysis and mitigation of biases in Greenland ice sheet mass balance trend estimates from GRACE mascon products, *Journal of Geophysical Research: Solid Earth*, 126, e2020JB020880, <https://doi.org/10.1029/2020JB020880>.
45. Zhang, J., **Liu, L.**, Su, L., and Che, T. (2021), Three in one: GPS-IR measurements of ground surface elevation changes, soil moisture, and snow depth at a permafrost site in the northeastern Qinghai-Tibet Plateau, *The Cryosphere*, 15, 3021–3033, <https://doi.org/10.5194/tc-15-3021-2021>.

46. Huang, L.[×], Liu, L., Luo, J., Lin, Z., and Niu, F. (2021), Automatically quantifying evolution of retrogressive thaw slumps in Beiluhe (Tibetan Plateau) from multi-temporal CubeSat images, *International Journal of Applied Earth Observations and Geoinformation*, 102, 102399, <https://doi.org/10.1016/j.jag.2021.102399>.
47. Xu, X., Liu, L., Schaefer, K., and Michaelides, R. (2021), Comparison of surface subsidence measured by airborne and satellite InSAR over permafrost areas near Yellowknife Canada, *Earth and Space Science*, 8, e2020EA001631, <https://doi.org/10.1029/2020EA001631>.
48. Michaelides, R. J., Chen, R. H., Zhao, Y., Schaefer, K., Parsekian, A. D., Sullivan, T., Moghaddam, M., Zebker, H. A., Liu, L., Xu, X., and Chen, J. (2021), Permafrost Dynamics Observatory (PDO) – Part I: Postprocessing and calibration methods of UAVSAR L-band InSAR data for seasonal subsidence estimation, *Earth and Space Science*, 8, e2020EA001630, <https://doi.org/10.1029/2020EA001630>.
49. Clayton, L. K., Schaefer, K., Battaglia, M. J., Bourgeau-Chavez, L., Chen, J., Chen, R. H., Chen, A., Bakian-Dogaheh, K., Grelik, S., Jafarov, E., Liu, L., Michaelides, R. J., Moghaddam, M., Parsekian, A., Rocha, A. V., Schaefer, S. R., Sullivan, T., Tabatabaenejad, A., Wang, W., Wilson, C., Zebker, H. A., Zhang, T., and Zhao, Y. (2021), Active layer thickness as a function of soil water content, *Environmental Research Letters*, 16, 055028, <https://doi.org/10.1088/1748-9326/abfa4c>.
50. Jiang, G., Liu, L., Barbour, A., Yang, H., and Lu, R. (2021), Physics-based evaluation of the maximum magnitude of potential earthquakes induced by the Hutubi (China) underground gas storage, *Journal of Geophysical Research: Solid Earth*, 126, e2020JB021379, <https://doi.org/10.1029/2020JB021379>.
51. Zhang, J., and Liu, L. (2021), Mining noise data for monitoring Arctic permafrost by using GNSS interferometric reflectometry, *Polar Science*, 29, 100649, <https://doi.org/10.1016/j.polar.2021.100649>.
52. Zhang, B., Yao, Y., Liu, L., and Yang, Y. (2021), Interannual ice mass variations over the Antarctic ice sheet from 2003 to 2017 were linked to El Niño-Southern Oscillation, *Earth and Planetary Science Letters*, 560, 116796, <https://doi.org/10.1016/j.epsl.2021.116796>.
53. Zhang, E., Liu, L., Huang, L.[×], and Ng, K. S.* (2021), An automated, generalized, deep-learning-based method for delineating the calving fronts of Greenland glaciers from multi-sensor remote sensing imagery, *Remote Sensing of Environment*, 254, 112265, <https://doi.org/10.1016/j.rse.2020.112265>.
54. Liu, W., Chen, X., Ran, J., Liu, L., Wang, Q., Xin, L., and Li, G. (2021), LaeNet: A novel lightweight multitask CNN for automatically extracting lake area and shoreline from remote sensing images, *Remote Sensing*, 13, 56, <https://doi.org/10.3390/rs13010056>.
55. Hu, Y., Liu, L., Wang, X., Zhao, L., Wu, T., Cai, J., Zhu, X. and Hao, J. (2021), Quantification of permafrost creep provides kinematic evidence for classifying a puzzling periglacial landform, *Earth Surface Processes and Landforms*, 46, 465–477, <https://doi.org/10.1002/esp.5039>.
56. Khan, S. A., Bjørk, A., Bamber, J., Morlighem, M., Bevis, M., Kjær, K., Mouginot, J., Løkkegaard, A., Holland, D., Aschwanden, A., Zhang, B., Helm, V., Korsgaard, N., Colgan, W., Larsen, N., Liu, L., Hansen, K., Barletta, V., Dahl-Jensen, T., Søndergaard, A., Csatho, B., Sasgen, I., Box, J., and Schenk, T. (2020), Centennial response of Greenland's three largest outlet glaciers, *Nature Communications*, 11, 5718, <https://doi.org/10.1038/s41467-020-19580-5>.
57. Zhang, B., Liu, L., Yao, Y., van Dam, T., and Khan, S. A. (2020), Improving the estimate of the secular variation of Greenland ice mass in the recent decades by incorporating a stochastic process, *Earth and Planetary Science Letters*, 549, 116518, <https://doi.org/10.1016/j.epsl.2020.116518>.
58. Zhang, J., Liu, L., and Hu, Y. (2020), Global Positioning System interferometric reflectometry (GPS-IR) measurements of ground surface elevation changes in permafrost areas in northern Canada, *The Cryosphere*, 14, 1875–1888, <https://doi.org/10.5194/tc-14-1875-2020>.
59. Huang, L., Luo, J., Lin, Z., Niu, F., and Liu, L. (2020), Using deep learning to map retrogressive thaw slumps in the Beiluhe region (Tibetan Plateau) from CubeSat images, *Remote Sensing of Environment*, 237, 111534, <https://doi.org/10.1016/j.rse.2019.111534>.
60. Jiang, G.[×], Qiao, X., Wang, X., Lu, R., Liu, L., Yang, H., Su, Y., Song, L., Wang, B., and Wong, T-f. (2020), GPS observed horizontal ground extension at the Hutubi (China) underground gas storage facility and its application to geomechanical modeling for induced seismicity, *Earth and Planetary Science Letters*, 530, 115943, <https://doi.org/10.1016/j.epsl.2019.115943>.

61. Huang, L., Baud, P., Cordonnier, B., Renard, F., **Liu, L.**, and Wong, T.-f. (2019), Synchrotron X-ray imaging in 4D: Multiscale failure and compaction localization in triaxially compressed porous limestone, *Earth and Planetary Science Letters*, 528, 115831, <https://doi.org/10.1016/j.epsl.2019.115831>.
62. Zhang, E., **Liu, L.**, and Huang, L. (2019), Automatically delineating the calving front of Jakobshavn Isbræ from multitemporal TerraSAR-X images: a deep learning approach, *The Cryosphere*, 13, 1729–1741, <https://doi.org/10.5194/tc-2019-14>.
63. Zhou, Z.[×], **Liu, L.**, Jiang, L., Feng, W., and Samsonov, S.V. (2019), Using long-term SAR backscatter data to monitor post-fire vegetation recovery in tundra environment, *Remote Sensing*, 11(19), 2230, <https://doi.org/10.3390/rs11192230>.
64. Zhang, B.[×], **Liu, L.**, Khan, S.A., van Dam, T., Bjørk, A.A., Peings, Y., Zhang, E., Bevis, M., Yao, Y., and Noël, B. (2019), Geodetic and model data reveal different spatio-temporal patterns of transient mass changes over Greenland from 2007 to 2017, *Earth and Planetary Science Letters*, 515, 154–163, <https://doi.org/10.1016/j.epsl.2019.03.028>.
65. Chen, X., **Liu, L.**, and Bartsch, A. (2019), Detecting soil freeze/thaw onsets in Alaska using SMAP and ASCAT data, *Remote Sensing of Environment*, 220, 59–70, <https://doi.org/10.1016/j.rse.2018.10.010>.
66. Wang, S., Xu, W., Xu, C., Yin, Z., Bürgmann, R., **Liu, L.**, and Jiang, G.[×] (2019), Changes in groundwater level possibly encourage shallow earthquakes in central Australia: The 2016 Petermann Ranges earthquake, *Geophysical Research Letters*, 46, 3189–3198, <https://doi.org/10.1029/2018GL080510>.
67. Michaelides, R. J., Schaefer, K., Zebker, H.A., Parsekian, A., **Liu, L.**, Chen, J., Natali, S., Ludwig, S., and Schaefer, S.R. (2019), Inference of the impact of wildfire on permafrost and active layer thickness in a discontinuous permafrost region using the remotely sensed active layer thickness (ReSALT) algorithm, *Environmental Research Letters*, 14(3), 035007, <https://doi.org/10.1088/1748-9326/aaf932>.
68. Huang, L., **Liu, L.**, Zhang, T., and Jiang, L. (2018), Automatic mapping of thermokarst landforms from remote sensing images using deep learning: A case study in the Northeastern Tibetan Plateau, *Remote Sensing*, 10(12), 2067, <https://doi.org/10.3390/rs10122067>.
69. Chen, J., **Liu, L.**, Zhang, T., Cao, B., and Lin, H. (2018), Using Persistent Scatterer Interferometry to map and quantify permafrost thaw subsidence: a case study of Eboling Mountain on the Qinghai-Tibet Plateau, *Journal of Geophysical Research: Earth Surface*, 123, 2663–2676, <https://doi.org/10.1029/2018JF004618>.
70. Chen, J., Günther, F., Grosse, G., **Liu, L.**, and Lin, H. (2018), Sentinel-1 InSAR measurements of elevation changes over Yedoma uplands on Sobo-Sise Island, Lena Delta, *Remote Sensing*, 10(7), 1152, <https://doi.org/10.3390/rs10071152>.
71. Ran, J.[×], Vizcaino, M., Ditmar, P., van den Broeke, M. R., Moon, T., Steger, C. R., Enderlin, E. M., Wouters, B., Noël, B., Reijmer, C. H., Klees, R., Zhong, M., **Liu, L.**, and Fettweis, X. (2018) Seasonal mass variations show timing and magnitude of meltwater storage in the Greenland ice sheet, *The Cryosphere*, 12, 2981–2999, <https://doi.org/10.5194/tc-12-2981-2018>.
72. Hu, Y.[◇], **Liu, L.**, Larson, K.M., Schaefer, K.M., Zhang, J., and Yao, Y. (2018), GPS Interferometric Reflectometry reveals cyclic elevation changes in thaw and freezing seasons in a permafrost area (Barrow, Alaska), *Geophysical Research Letters*, 45, 5581–5589, <https://doi.org/10.1029/2018GL077960>.
73. **Liu, L.** and Larson, K.M. (2018), Decadal changes of surface elevation over permafrost area estimated using reflected GPS signals, *The Cryosphere*, 12, 477–489, <https://doi.org/10.5194/tc-12-477-2018>.
74. Zhang, B.[×], Zhang, E., **Liu, L.**, Khan, S.A., van Dam, T., Yao, Y., Bevis, M., and Helm, V. (2018), Geodetic measurements reveal short-term changes of glacial mass near Jakobshavn Isbræ (Greenland) from 2007 to 2017, *Earth and Planetary Science Letters*, 503, 216–226, <https://doi.org/10.1016/j.epsl.2018.09.029>.
75. Schuster, P. F., Schaefer, K.M., Aiken, G.R., Antweiler, R.C., Dewild, J.F., Gryziec, J.D., Gusmeroli, A., Hugelius, G., Jafarov, E., Krabbenhoft, D.P., **Liu, L.**, Herman-Mercer, N., Mu, C., Roth, D. A., Schaefer, T., Striegl, R. G., Wickland, K. P., and Zhang, T. (2018), Permafrost stores a globally significant amount of mercury, *Geophysical Research Letters*, 45, 1463–1471, <https://doi.org/10.1002/2017GL075571>. 🏆 **Top-cited Paper**
76. Wu, Z.[◇], Zhao, L., **Liu, L.**, Zhu, R., Gao, Z., Qiao, Y., Tian, L., Zhou, H., and Xie, M. (2018), Surface deformation

- monitoring in the permafrost regions over Tibetan Plateau using Sentinel-1 data, *Sciences in Cold and Arid Regions*, 10(2), 114–125, <https://doi.org/10.3724/SP.J.1226.2018.00114>.
77. Jafarov E. E., Parsekian, A., Schaefer, K., **Liu, L.**, Chen, A.C., Panda, S.K., and Zhang, T. (2018), Estimating active layer thickness and volumetric water content from ground penetrating radar measurements in Barrow, Alaska, *Geoscience Data Journal*, 4, 72–79, <https://doi.org/10.1002/gdj3.49>.
 78. Zhang, B.[×], **Liu, L.**, Khan, S. A., van Dam, T., Zhang, E., and Yao, Y. (2017), Transient variations in glacial mass near Upernavik Isstrøm (west Greenland) detected by the combined use of GPS and GRACE data, *Journal of Geophysical Research: Solid Earth*, 122, 10626–10642, <https://doi.org/10.1002/2017JB014529>.
 79. **Liu, L.**, Khan, S. A., van Dam, T., Ma, J. H. Y., and Bevis, M. (2017), Annual variations in GPS-measured vertical displacements near Upernavik Isstrøm (Greenland) and contributions from surface mass loading, *Journal of Geophysical Research: Solid Earth*, 122, 677–691, <https://doi.org/10.1002/2016JB013494>.
 80. Wang, X.[◇], **Liu, L.**, Zhao, L., Wu, T., Li, Z., and Liu, G. (2017), Mapping and inventorying active rock glaciers in the northern Tien Shan of China using satellite SAR interferometry, *The Cryosphere*, 11, 997–1014, <https://doi.org/10.5194/tc-11-997-2017>.
 81. Iwahana G., Uchida, M., **Liu, L.**, Gong, W., Meyer, F.J., Guritz, R., Yamanokuchi, T., and Hinzman, L., (2016), Field evidence and InSAR detection of thermokarst after a tundra wildfire, using ALOS-PALSAR, *Remote Sensing*, 8(3), 218, <https://doi.org/10.3390/rs8030218>.
 82. Chen, A., Parsekian, A., Schaefer, K., Jafarov, E., Panda, S., **Liu, L.**, Zhang, T., and Zebker, H. (2016), Ground-penetrating radar-derived measurements of active-layer thickness on the landscape scale with sparse calibration at Toolik and Happy Valley, Alaska, *Geophysics*, 81(2), H1–H11, <https://doi.org/10.1190/geo2015-0124.1>.
 83. **Liu, L.**, Schaefer, K.M., Chen, A.C., Gusmeroli, A., Zebker, H.A., and Zhang, T. (2015), Remote sensing measurements of thermokarst subsidence using InSAR, *Journal of Geophysical Research: Earth Surface*, 120, 1935–1948, <https://doi.org/10.1002/2015JF003599>.
 84. Jones, B. M., Grosse, G., Arp, C.D., Miller, E., **Liu, L.**, Hayes, D.J., and Larsen, C.F. (2015), Recent Arctic tundra fire initiates widespread thermokarst development, *Scientific Reports*, 5, 15865, <https://doi.org/10.1038/srep15865>.
 85. Schaefer, K., **Liu, L.**, Parsekian, A., Jafarov, E., Chen, A., Zhang, T., Gusmeroli, A., Panda, S., Zebker, H.A., and Schaefer, T. (2015), Remotely Sensed Active Layer Thickness (ReSALT) at Barrow, Alaska using Interferometric Synthetic Aperture Radar, *Remote Sensing*, 7, 3735–3759, <https://doi.org/10.3390/rs70403735>.
 86. Gusmeroli, A., **Liu, L.**, Schaefer, K., Zhang, T., Schaefer, T., and Grosse, G. (2015), Active layer stratigraphy and organic layer thickness at a thermokarst site in Arctic Alaska identified using Ground Penetrating Radar, *Arctic Antarctic and Alpine Research*, 47(2), 195–202, <https://doi.org/10.1657/AAAR00C-13-301>.
 87. **Liu, L.**, Jafarov, E.E., Schaefer, K.M., Jones, B.M., Zebker, H.A., Williams, C.A., Rogan, J., and Zhang, T. (2014), InSAR detects increase in surface subsidence caused by an Arctic tundra fire, *Geophysical Research Letters*, 41, 3906–3913, <https://doi.org/10.1002/2014GL060533>.
 88. **Liu, L.**, Schaefer, K., Gusmeroli, A., Grosse, G., Jones, B. M., Zhang, T., Parsekian, A. D., and Zebker, H. A. (2014), Seasonal thaw settlement at drained thermokarst lake basins, Arctic Alaska, *The Cryosphere*, 8, 815–826, <https://doi.org/10.5194/tc-8-815-2014>.
 89. Khan, S. A., Kjær, K., Bevis, M., Bamber, J. L., Wahr, J., Kjeldsen, K., Bjørk, A., Korsgaard, N., Stearns, L., van den Broeke, M. R., **Liu, L.**, Larsen, N., and Muresan, I. (2014), Sustained mass loss of the Northeast Greenland ice sheet triggered by regional warming, *Nature Climate Change*, 4, 292–299, <https://doi.org/10.1038/nclimate2161>.
 90. Mu, C., Zhang, T., Schuster, P.F., Schaefer, K., Wickland, K.P., Repert, D.A., **Liu, L.**, Schaefer, T., and Cheng, G. (2014), Carbon and geochemical properties of cryosols on the North Slope of Alaska, *Cold Regions Science and Technology*, 100, 59–67, <https://doi.org/10.1016/j.coldregions.2014.01.001>.
 91. **Liu, L.**, Millar, C.I., Westfall, R.D., and Zebker, H.A. (2013), Surface motion of active rock glaciers in the Sierra Nevada, California, USA: inventory and a case study using InSAR, *The Cryosphere*, 7, 1109–1119, <https://doi.org/10.5194/tc-7-1109-2013>.

92. Parsekian, A., Grosse, G., Walbrecker, J.O., Müller-Petke M., Keating, K., **Liu, L.**, Jones, B.M., and Knight, R. (2013), Detecting unfrozen sediments below thermokarst lakes with surface nuclear magnetic resonance, *Geophysical Research Letters*, 40, 535–540, <https://doi.org/10.1002/grl.50137>.
93. Wahr, J., Khan, S.A., van Dam, T., **Liu, L.**, van Angelen, J.H., van den Broeke, M.R., and Meertens, C.M. (2013), The use of GPS horizontals for loading studies, with applications to northern California and southeast Greenland, *Journal of Geophysical Research: Solid Earth*, 118, 1795–1806, <https://doi.org/10.1002/jgrb.50104>.
94. Nielsen, K., Khan, S.A., Spada, G., Wahr, J., Bevis, M., **Liu, L.**, and van Dam, T. (2013), Vertical and horizontal surface displacements near Jakobshavn Isbræ driven by melt-induced and dynamic ice loss, *Journal of Geophysical Research: Solid Earth*, 118, 1837–1844, <https://doi.org/10.1002/jgrb.50145>.
95. **Liu, L.**, Wahr, J., Howat, I., Khan, S.A., Joughin, I., and Furuya, M. (2012), Constraining ice mass loss from Jakobshavn Isbræ (Greenland) using InSAR-measured crustal uplift, *Geophysical Journal International*, 188: 994–1006, <https://doi.org/10.1111/j.1365-246X.2011.05317.x>.
96. **Liu, L.**, Schaefer, K., Zhang, T., and Wahr, J. (2012), Estimating 1992–2000 average active layer thickness on the Alaskan North Slope from remotely sensed surface subsidence, *Journal of Geophysical Research: Earth Surface*, 117, F01005, <https://doi.org/10.1029/2011JF002041>.
97. **Liu, L.**, Zhang, T., and Wahr, J. (2010), InSAR measurements of surface deformation over permafrost on the North Slope of Alaska, *Journal of Geophysical Research: Earth Surface*, 115, F03023, <https://doi.org/10.1029/2009JF001547>.
98. Khan, S. A., **Liu, L.**, Wahr, J., Howat, I., Joughin, I., van Dam, T., and Fleming, K. (2010), GPS measurements of crustal uplift near Jakobshavn Isbræ due to glacial ice mass loss, *Journal of Geophysical Research: Solid Earth*, 115, B09405, <https://doi.org/10.1029/2010JB007490>.
99. Shen W., **Liu, L.**, and Ning, J. (2007), The inner core's super rotation and its influences on the gravity field, *Chinese Journal of Geophysics* (in Chinese), 50(2), 430–436.

Book Chapters

100. van Dam, T., Whitehouse, P., and **Liu, L.** (2024), GNSS and the cryosphere, in *GNSS Monitoring of the Terrestrial Environment: Earthquakes, Volcanoes, and Climate Changes*, C. Kreemer and Y. Aoki, Elsevier. <https://doi.org/10.1016/B978-0-323-95507-2.00015-3>.
101. **Liu, L.** (2021), A review of deep learning for cryospheric studies, in *Deep Learning for the Earth Sciences*, G. Camps-Valls, D. Tuia, X. Zhu, and M. Reichstein, Eds., Wiley & Sons. <https://doi.org/10.1002/9781119646181.ch17>.

Papers Under Review

102. Sun, Z., Bolch, T., Falaschi, D., Bhattacharya, A., and Liu, L., Multi-decadal evolution of ice-debris complexes at Muztagh Ata (Eastern Pamir) derived from Corona KH-4A and Pléiades stereo images, *EGU sphere*, <https://doi.org/10.5194/2026-3678>, 2026.
103. Huang, L., **Liu, L.**, Xia, Z., Ma, P., Kwan, M., Locating retrogressive thaw slumps from multi-source and multi-temporal remote sensing data using AI foundation models, submitted to *Permafrost and Periglacial Processes*.
104. Moor, M., et al., Soil moisture and active layer thickness across Alaska USA and Northwestern Canada, submitted to *Scientific Data*.

Open-access Codes and Data Products (selected)

105. Fan, C., and **Liu, L.** (2024). FanInSAR: A Fancy InSAR time series library, in a Pythonic, fast, and flexible way (0.0.1). Zenodo. <https://doi.org/10.5281/zenodo.11398347>
106. Sun, Z., Hu, Y., Racoviteanu, A., **Liu, L.**, Harrison, S., Wang, X., Cai, J., Guo, X., He, Y., and Yuan, H. (2024), TPRoGI: a comprehensive rock glacier inventory for the Tibetan Plateau using deep learning (1.0). Zenodo, <https://doi.org/10.5281/zenodo.10732042>.
107. Xia, Z., **Liu, L.**, Mu, C., Peng, X., Zhao, Z., Huang, L., Luo, J., and Fan, C. (2024), Annual inventories of retrogressive thaw slumps across the Qinghai-Tibet Plateau from 2016 to 2022. Zenodo, <https://doi.org/10.5281/zenodo.10928346>.

108. Zhao, Z., Fan, C., **Liu, L.** (2023), Geo SAM: A QGIS plugin using Segment Anything Model (SAM) to accelerate geospatial image segmentation (1.1.0). Zenodo, <https://doi.org/10.5281/zenodo.8191039>.
109. Xia, Z., Huang, L., **Liu, L.** (2021), An Updated Inventory of Retrogressive Thaw Slumps Along the Vulnerable Qinghai-Tibet Engineering Corridor. Zenodo, <https://doi.org/10.5281/zenodo.6397029>.
110. Schaefer, K., Michaelides, R., Chen, R., Sullivan, T., Parsekian, A., Zhao, Y., Bakian-Dogaheh, K., Tabatabaenejad, A., Moghaddam, M., Chen, J., Chen, A., **Liu, L.**, and Zebker, H. (2021), ABoVE: Active Layer Thickness Derived from Airborne L- and P-band SAR, Alaska, 2017. ORNL DAAC, Oak Ridge, Tennessee, USA. <https://doi.org/10.3334/ORNLDAAAC/1796>.
111. Zhang, J. and **Liu, L.** (2020), Reflector heights in the Arctic permafrost areas measured by GNSS interferometric reflectometry. Zenodo, <https://doi.org/10.5281/zenodo.4319372>.
112. Zhang, J., **Liu, L.**, Y. Hu (2019), Reflector heights measured by GPS-IR at Alert, Resolute Bay, Repulse Bay, Baker Lake, and Iqaluit in northern Canada. PANGAEA, <https://doi.org/10.1594/PANGAEA.904347>.
113. **Liu, L.**, K. M. Larson (2018), Surface elevation changes near Barrow (Alaska) measured using reflected GPS signals. PANGAEA, <https://doi.pangaea.de/10.1594/PANGAEA.885935>.
114. Hu, Y., **Liu, L.**, K. M. Larson (2018), The decadal reflector heights for SG27 in Barrow, Alaska (2007-2016). PANGAEA, <https://doi.pangaea.de/10.1594/PANGAEA.884941>.
115. Schaefer, K., R.J. Michaelides, R.H. Chen, T. Sullivan, A.D. Parsekian, K. Bakian-dogaheh, A. Tabatabaenejad, M. Moghaddam, J. Chen, A.C. Chen, **Liu, L.** and H.A. Zebker (2019), ABoVE: Active Layer Thickness Derived from Airborne L- and P-band SAR, Alaska, 2017. ORNL DAAC, Oak Ridge, Tennessee, USA. <https://doi.org/10.3334/ORNLDAAAC/1676>.
116. Zhang, B., **Liu, L.**, S. A. Khan, T. van Dam, E. Zhang, and Y. Yao (2017), GPS and GRACE inferred uplifts and extracted transient and seasonal signals due to glacial mass change near Upernavik Isstrm, PANGAEA, <https://doi.org/10.1594/PANGAEA.880159>.
117. Wang, X., **Liu, L.**, L. Zhao, T. Wu, Z. Li, and G. Liu (2017), An inventory of active rock glaciers in the northern Tien Shan of China compiled using satellite SAR interferometry, PANGAEA, <https://doi.pangaea.de/10.1594/PANGAEA.874616>.
118. Jafarov, E., A. Parsekian, K. Schaefer, **Liu, L.**, A. Chen, S.K. Panda, and T. Zhang (2016), Pre-ABoVE: Active Layer Thickness and Soil Water Content, Barrow, Alaska, 2013. ORNL DAAC, Oak Ridge, Tennessee, USA. <http://dx.doi.org/10.3334/ORNLDAAAC/1355>.
119. Chen, A., A. Parsekian, K. Schaefer, E. Jafarov, S.K. Panda, **Liu, L.**, T. Zhang, and H.A. Zebker. 2015. Pre-ABoVE: Ground-penetrating Radar Measurements of ALT on the Alaska North Slope. ORNL DAAC, Oak Ridge, Tennessee, USA. <http://dx.doi.org/10.3334/ORNLDAAAC/1265>.
120. **Liu, L.**, K. Schaefer, A. Chen, A. Gusmeroli, E. Jafarov, S. Panda, A. Parsekian, T. Schaefer, H. A. Zebker, T. Zhang. 2015. Pre-ABoVE: Remotely Sensed Active Layer Thickness, Barrow, Alaska, 2006-2011. ORNL DAAC, Oak Ridge, Tennessee, USA. <http://dx.doi.org/10.3334/ORNLDAAAC/1266>.
121. **Liu, L.**, K. Schaefer, A. Chen, A. Gusmeroli, E. Jafarov, S. Panda, A. Parsekian, T. Schaefer, H. A. Zebker, T. Zhang. 2015. Pre-ABoVE: Remotely Sensed Active Layer Thickness, Prudhoe Bay, Alaska, 1992-2000. ORNL DAAC, Oak Ridge, Tennessee, USA. <http://dx.doi.org/10.3334/ORNLDAAAC/1267>.

Other Publications (non-peer-reviewed)

122. Wig, E., Meng, T., Schaefer, K., Chen, J., Chen, R., French, N., Hoy, E., **Liu, L.**, Chorsi, T., Siqueira, P., Zebker, H., Zwieback, S., and Michaelides, R. (2026), Repeat-Pass InSAR Mission Needs and Priorities for Permafrost, White Paper for the Decadal Survey for Earth Science and Applications from Space 2028–2037.
123. Feng, W., Xiong, Y., Yi, S., Zhong, B., Chen, X., Zhong, Y., Pan, Y., **Liu, L.**, Wang, W., and Zhong, M. (2023), Recent Progress on Hydrogeodesy in China, *Journal of Geodesy and Geoinformation Science*, 6(3), 124–134, <https://doi.org/10.11947/j.JGGS.2023.0312>.
124. Panda, S., K. Schaefer, **Liu, L.**, E. Jafarov, A. Parsekian, and A. Chen (2015), Connecting lake area change, ground subsidence and permafrost carbon dynamics in Prudhoe Bay, *Changing Ice: A Newsletter of Cryosphere Research in Alaska*, December 2015.

125. **Liu, L.** (2015) Melting Glaciers in High Asia and their Impacts on Water Sustainability, *CUHK Sustainable Campus*, No 10, October 2015.
126. Contributing author to Bartsch et al. (2014) Requirements for Monitoring of Permafrost in Polar Regions: A community white paper in response to the WMO Polar Space Task Group.
127. **Liu, L.**, C. Millar, R. Westfall, and H. Zebker (2013), Taking a Census of California Rock Glaciers from Space, *Mountain Views*, Volume 7, No 2, November 2013.
128. **Liu, L.** (2011), Studying changes in the cryosphere using radar interferometry: permafrost surface subsidence and glacial unloading deformation, *PhD thesis*, University of Colorado.
129. **Liu, L.**, T. Zhang, K. Schaefer, and J. Wahr, InSAR Observations Revealed Surface Subsidence Over Permafrost in Northern Alaska, *Alaska Satellite Facility News and Notes*, 2011 Spring Volume.

RECENT INVITED TALKS

- *Uncovering controlling factors on rock glacier velocities using Explainable AI*, 3rd Chinese Cryosphere Remote Sensing Symposium 2026
- *Vegetation recovery following thaw slump disturbance*, Peking University and Chinese Academy of Sciences 2026
- *Rock glacier distribution and movement velocities on the Tibetan Plateau*, Peking University 2026
- *Tracking Greenland meltwater storage through mass loading deformation*, ISTerre 2025
- *Monitoring permafrost degradation using InSAR*, Southern University of Science and Technology 2025
- *Deep learning for Cryospheric Studies*, Lanzhou University of Technology 2025
- *Tracking Greenland meltwater storage through mass loading deformation*, Wuhan University 2025
- *Monitoring permafrost degradation using space geodesy*, China University of Geosciences (Wuhan) 2025
- *Spatial-temporal evolution of thaw slumps on the Qinghai-Tibetan Plateau*, Innovation Academy for Precision Measurement Science and Technology, Chinese Academy of Sciences 2025
- *Investigating frozen ground dynamics using GNSS Interferometric Reflectometry*, GNSS-IR Webinar Series 2025
- *GNSS for Ice: Tracking Greenland meltwater storage through mass loading deformation*, University of Utah 2024
- *Geodetic studies of frozen ground*, Michigan State University 2024
- *Geophysical investigations of thawing permafrost*, University of Utah 2024
- *Geodesy for permafrost Essential Climate Variable*, Sun Yat-sen University 2024
- *Remote sensing of frozen ground*, University of Hong Kong 2023
- *Deep learning for Cryospheric Studies*, Lanzhou University 2023
- *Environmental, socioeconomic, and health risks of thawing permafrost in Tibet*, Tongji University 2023
- *Geophysical investigations of thawing permafrost*, University of Texas at Austin 2022
- *Geodetic and remote sensing observations of thawing permafrost*, Hong Kong Polytechnic University 2022
- *Geodetic studies of frozen ground*, Peking University 2022
- *Deep learning for earth sciences*, Geological Society of London Hong Kong Regional Group 2021
- *Geodetic studies of frozen ground*, AGU Geodesy webinar 2021
- *AI for Cryosphere*, University of Science and Technology of China 2021

RESEARCH STUDENTS SUPERVISED

CUHK Postgraduates

- **Joseph H.Y. Ma** MPhil, 2016
Understanding temporal changes of glacial dynamics with numerical modeling: A case study of Upernavik Isstrøm, Greenland
After CUHK: PhD at National University of Singapore, Now: Data Scientist in industry
- **Lingcao Huang** PhD, 2019
Mapping non-lake thermokarst landforms on the Tibetan Plateau using remote sensing and deep learning
Now: Research Assistant Professor at CUHK
- **Jie Chen** PhD, 2019 (co-supervised with Prof. Hui Lin)

Studying permafrost and active layer dynamics in Tibet and Arctic by multi-temporal radar interferometry
Now: Postdoc at University of Alaska Fairbanks

- **Enze Zhang** PhD, 2020
Investigating front variations of Greenland glaciers using multi-temporal remote sensing images & deep learning
Now: Research Assistant Professor at Hong Kong University of Science and Technology
- **Xiyu Chen** PhD, 2021
Detecting landscape freeze/thaw onsets and states using active and passive microwave remote sensing data
Now: Chongqing Forestry Research Institute
- **Yan Hu** PhD, 2021
Rock glacier kinematics: A proxy for assessing periglacial dynamics & ground ice content on the Tibetan Plateau
Now: Postdoc at University of Fribourg
- **Jiahua Zhang** PhD, 2021
Investigating frozen ground dynamics by using Global Navigation Satellite System interferometric reflectometry
Now: Project Scientist at University Corporation for Atmospheric Research (UCAR)
- **Billy Ho Ming Tsang** MPhil, 2022
Landslide mapping from remote sensing images using deep learning: a case study in Lantau Island, Hong Kong
Now: PhD student at University of Hong Kong
- **Xingyu (Carol) Xu** PhD, 2023
Investigating Glacial Lakes in the Hindu Kush Karakoram Himalaya Region Using Multi-Source Remote Sensing & Deep Learning
Now: Postdoc at Wuhan University
- **Zhuoxuan (Summer) Xia** PhD, 2025
Investigating the spatial distribution, dynamics, and vegetation recovery of retrogressive thaw slumps on the Qinghai-Tibetan Plateau
Now: Postdoc at University of Illinois
- **Zhangyu (Joe) Sun** PhD, 2025
A comprehensive study of rock glacier distribution, velocities, and water storage in High Mountain Asia
Now: Tenure-track faculty member at Hohai University
- **Jianlong (Herbert) He** PhD, 2022–present
- **Wensong (Bill) Zhang** PhD, 2023–present
- **Mengze Li** PhD, 2024–present
- **Saira Parveen** PhD, 2024–present (co-supervised with Prof. Alex Chow)
- **Luming (Louis) Yang** PhD, 2025–present
- **Yifan Bu** PhD, 2025–present
- **Marcellus Leung Yu Him** MPhil, 2025–present (co-supervised with Prof. Martin Tsui)
- **Wei Wu** PhD, 2026–present
- **Lin Peng** PhD, 2026–present, University of Exeter (co-supervised with Profs. Stephan Harrison and Karen Anderson)
- **Ruixin (Rae) Wang** PhD, 2026–present
- **Yujie Sun** PhD, 2026–present

Postdoctoral Fellows: Bo Hu (2015), Zhiwei Zhou (2016–2017), Bao Zhang (2017–2018), Guoyan Jiang (2016–2019, co-supervised with T-f. Wong), Jiangjun Ran (2018), Lingcao Huang (2020), Yan Hu (2021–2024), Xiaofan Zhu (2022), Zhuoyi (Joey) Zhao (2022–2023), Xingyu Xu (2023–2025)

Visting Students: Enze Zhang (USTC, 2015), Weiyu Zheng (USTC, 2016), Xiaowen Wang (SWJTU, 2016–2017), Wanwan Shao (Lanzhou U, 2016), Zhenming Wu (CAS, 2017), Jiahui Wang (USTC, 2017), Yongxin Liu (Wuhan U, 2017), Yufeng Hu (Wuhan U, 2017–2018), Weifan Zhou (Jilin U, 2018), Linyang Xin (Wuhan U, 2019), Yidan Ding

(Jilin U, 2019), Haoran Wang (Jilin U, 2019), Chengyan Fan (Lanzhou U, 2021–2022), Mingkai Chen (East China University of Technology, 2023), Tian Liang (Southwest Jiaotong University, 2025)

SERVICE

University, College, and Department Services at CUHK

- Member, AISCI MSc Programme Committee, Faculty of Science 2026–present
- Member, Programme Committee, MSc in Earth and Environmental Analytics for Sustainability, Department of Earth and Environmental Sciences 2022–present
- Member, Executive committee, Department of Earth and Environmental Sciences 2022–2026
- Department coordinator, Chung Chi College 2015–present
- Member, Expert Committee of the Jockey Club Museum of Climate Change 2019–present
- Head, Graduate Division of Earth and Environmental Sciences 2020–2025
- Member, Search committee of Research Assistant Professor, Dept. of Earth & Environ. Sciences 2024
- Member, Faculty of Science Working Group for Earth & Environmental Sciences Programme 2021–2022
- Chair, Search committee of Ng Yin Ying Assistant Professor, Earth System Science Programme 2021
- Member, Graduate committee, Graduate Division of Earth and Atmospheric Sciences 2014–2024
- Institutional representative, UNAVCO and WInSAR 2015–2023
- Member, Search committee of assistant and associate professors, Earth System Science Programme 2018
- Member, Review panel of Master of Science in GeoInformation Science 2017
- Member, Lecturer search committee, Earth System Science Programme 2015
- Library committee, Earth System Science Programme 2014

Editorship

- Editor, *Earth Observation* 2026–present
- Editor, *The Cryosphere* 2026–present
- Science Advisor, *Eos* (AGU's science news magazine) 2024–present
- Member, Editorial Board, *Cryospheric Science and Engineering* 2026–present
- Member, Editorial Board, *The Innovation Insights* 2026–present
- Member, Editorial Board, *Journal of Cold Regions Engineering* 2023–present
- Member, Editorial Board, *Geodesy and Geodynamics* 2022–present
- Guest Editor, *Remote Sensing* 2022–2023

Member of International Committees or Groups

- AGU Geodesy Committee 2024–present
- AGU Geodesy Awards Committee 2023–present
- Advisory Board, International Permafrost Association (IPA) Standing Committee for Rock Glacier Inventories and Kinematics Executive Committee 2023–present
- Joint Work Group 'Geodesy for the Cryosphere: advancing the use of geodetic data in polar climate modelling', IAG Inter-Commission Committee on Geodesy for Climate Research 2020–present
- Director for Students, IEEE Geoscience and Remote Sensing Society (GRSS) Hong Kong Chapter 2024–present
- Treasurer, IEEE GRSS Hong Kong Chapter 2022–2024
- Vulnerability of Permafrost Carbon Research Coordination Network 2012–present
- IPA Action Group 'Retrogressive thaw slump inventory' 2022–2024
- IPA Action Group 'Rock Glacier Inventories and Kinematics' 2018–2023
- IPA Action Group 'Towards a Permafrost Thaw Subsidence Product within the GTN-P database' 2018–2020

Reviewer: *Earth and Planetary Science Letters, Earth Surface Processes and Landforms, Earth System Science Data, Frontiers in Earth Science, GISScience & Remote Sensing Geomorphology, Geophysical Research Letters, GPS Solutions, IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, IEEE Transactions on Geoscience and Remote Sensing, International Journal of Digital Earth, ISPRS Journal of Photogrammetry and Remote Sensing, Journal of Geodesy, Journal of Geophysical Research, Journal of Hydrology, Permafrost and Periglacial Processes, Pure and Applied Geophysics, Remote Sensing of Environment, Science Advances, Scientific Reports, Soil Science Society of America Journal, The Cryosphere, etc.*

IPCC's Special Report on Ocean and Cryosphere in a Changing Climate, US National Science Foundation, Netherlands Space Office, Hong Kong Research Grants Council, etc.

Conference Chair or Session Convener

- Permafrost in Transition, JpGU-AGU Joint Meeting 2026
- 2nd Chinese Symposium on InSAR 2026
- International Symposium on Polar Sciences – The 40th Anniversary of CHINARE 2024
- 1st Chinese Symposium on InSAR 2024
- Co-Chair, International Conference on Earth, Energy and Environmental Sciences for Carbon Neutrality 2023
- Co-Chair, The 2nd EAGE/SEG Workshop on Geophysical Aspects of Smart Cities 2022
- 5th International Workshop on Rock Physics 2019
- 2nd international conference of Digital Belt and Road (DBAR 2017) & 3rd international conference on remote sensing applications in tropical and subtropical areas (RSATSA 2017) 2017
- 3rd International Conference on Sensors and Models in Photogrammetry and Remote Sensing 2015
- 8th World Chinese Geosciences Congress 2015
- Recent Advances in SAR Technology for Earth Observation AGU Meeting 2016
- Scientific Exploration of the Earth with Multi-modal Remote Sensing Mission AGU Meeting 2015
- Advances in InSAR Data Processing for Earth System Applications AGU Meeting 2014
- Advances in Geophysical Characterization of Permafrost Systems AGU Meeting 2013

Member of Thesis Committee

2025–2026

Yi Zheng, PhD, CUHK (Chair)
 Lin Kong, PhD, CUHK (Chair)
 CHAN Sheung Ki Suki, MPhil, CUHK (Chair)
 LAU Chung Shing Martin, MPhil, CUHK (Chair)
 CHENG Wing Kok Kylie, MPhil, CUHK (Chair)
 Donger Lai, PhD, CUHK (Chair)
 Feier Yan, PhD, HKUST
 HO Chung Yan Joanne, MPhil, CUHK (Chair)
 LAW Chak Hay Hayden, MPhil, CUHK (Chair)
 Yukai Hu, MPhil, CUHK (Chair)
 Zilin Song, PhD, CUHK (Chair)

2022–2023

Chenxi Hu, PhD, CUHK (Chair)
 Cong Liu, PhD, University of Hong Kong
 Dingrui Cao, PhD, CUHK (Chair)
 Hemraj Bhattarai, PhD, CUHK (Chair)
 CHOW Tsun Ngai, PhD, CUHK (Chair)
 Han Chen, PhD, CUHK (Chair)
 Suli Yao, PhD, CUHK (Chair)
 Xiaolu Jiang, PhD, CUHK (Chair)

2020

Gaohua Zhu, PhD, CUHK
 LAM Hoi Ki, MPhil, CUHK (Chair)
 Pengcheng Zhou, PhD, CUHK
 Shihan Sun, PhD, CUHK (Chair)
 WONG Yat Chun, MPhil, CUHK (Chair)
 Xingfu Li, MPhil, CUHK
 Xueying Liu, PhD, CUHK (Chair)

2024

Adnan Barkat, PhD, CUHK (Chair)
 CHU Wai Ching, MPhil, CUHK (Chair)
 Jinping Zi, PhD, CUHK (Chair)
 Junhao Song, PhD, CUHK (Chair)
 LAU Tsz Lam, MPhil, CUHK (Chair)
 NG Sze In Madeleine, PhD, CUHK (Chair)
 Tiangang Yuan, PhD, CUHK (Chair)
 Yiling Zheng, PhD, CUHK (Chair)
 Zherong Wu, PhD, CUHK

2021

NG Tsin Hung, MPhil, CUHK (Chair)
 Rui Zhao, PhD, CUHK (Chair)
 TAO Ka Chuen, MPhil, CUHK (Chair)
 TAM Hiu Fai, MPhil, CUHK (Chair)
 Wei Lun Alan LIM, PhD, CUHK (Chair)
 Xiang Chen, PhD, CUHK (Chair)
 Yi Luan, PhD, CUHK
 Zhouyi Zhao, PhD, CUHK

2017–2019

FUNG Ka Ming, PhD, CUHK (Chair)
 LEE Hing Bun, Martin, MPhil, CUHK
 Bing He, MPhil, CUHK
 Keren Dai, PhD, Southwest Jiaotong University
 Tanghua Li, PhD, University of Hong Kong
 Xiaowen Wang, PhD, Southwest Jiaotong University

Outreach

- Interviews with local and national media on polar studies: *Headline Daily*, *Hong Kong Economic Journal*, *i-CABLE News*, *Radio and Television Hong Kong*, *Sing Tao Daily*, *Wen Wei Po* 2014–present
- Public talks on polar sciences and climate change 2014–present
 Malvern College Hong Kong
 Hong Kong Regional Group of The Geological Society London

Geological Society of Hong Kong
CUHK Sustainable Development Goals Forum
Hong Kong Jockey Club Museum of Climate Change
13th Lau Oi Wah Memorial Science Lecture Series, CUHK
CUHK Knowledge Enrichment Programme for Secondary School Students